



Teacher Copy — includes the Answer Key & Teacher Guide. Do not distribute to students.

UNIT 10 • STANDARD 6.GR.2

Volume with Whole Number Edges

Lesson 10-1 • Teacher Copy

Name: _____

Date: _____

Class: _____



TODAY'S OBJECTIVES

Content Objective: I can find the volume of a rectangular prism with whole-number edges using $\text{length} \times \text{width} \times \text{height}$.

Language Objective: I can explain my work using the words volume, rectangular prism, cubic units, and edge.



See **Learn It** for the explanation and worked examples, and the **Vocab** tab for the words. These are your notes to fill in and keep.



Fill in each blank as we go. Use the Word Bank to help you.



WORD BANK — FILL EACH BLANK WITH THE BEST WORD

Volume

Rectangular prism

Cubic units

Length, width, height

Net

Edge



Tap any word to see what it means and a picture.

1

The amount of space inside a three-dimensional solid is its

2

A solid with six rectangular faces, like a box, is a

3

The unit used to measure volume, like cubic centimeters, is

4

The three edge measurements of a rectangular prism are its

5

A flat pattern that folds up into a solid is a

6

A line segment where two faces of a solid meet is an

Reflect — Exit Ticket

A rectangular prism has $l = 11$ in, $w = 5$ in, $h = 4$ in. What is the volume?

- A. 220 in^3
- B. 55 in^3
- C. 220 in^2
- D. 200 in^3

Your answer:

Answer Key & Teacher Guide

1. **Notes 1:** Volume
2. **Notes 2:** Rectangular prism
3. **Notes 3:** Cubic units
4. **Notes 4:** Length, width, height
5. **Notes 5:** Net
6. **Notes 6:** Edge
7. **Watch for this mistake:** *A common mistake in Volume with Whole Number Edges is skipping the key idea: "To find the volume of a box, multiply all three edges: $V = \text{length} \times \text{width} \times \text{height}$." — always check your work against this rule before you submit.*
8. **Try It 1:** A. 30 cubic units — $V = \text{length} \times \text{width} \times \text{height} = 5 \times 3 \times 2 = 30$ cubic units.
9. **Try It 2:** A. 24 cubic units — *Each layer has 6 cubes and there are 4 layers, so $6 \times 4 = 24$ cubic units. This is the same as $\text{length} \times \text{width} \times \text{height}$ with one factor being the number of layers.*
10. **Exit Ticket:** A. 220 in^3 — $V = 11 \times 5 \times 4 = 220$ cubic inches. Remember: volume uses cubic units (in^3).